

Independent Replication Report:
*The Hidden Subgroup Problem and Eigenvalue Estimation on a
Quantum Computer*
(Mosca & Ekert, 1998; arXiv:quant-ph/9903071)

Independent replication (QC-200 wave, 2026-07-05)

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Abstract

We reproduce the central computational primitives of Mosca & Ekert’s 1998 paper, which unifies the abelian Hidden Subgroup Problem (HSP) with Kitaev’s eigenvalue estimation via a common quantum Fourier / phase-kickback framework. We (i) implement Quantum Phase Estimation (QPE) for a single-qubit unitary $U = \text{diag}(1, e^{2\pi i\varphi})$ at $\varphi \in \{1/3, 1/4, 1/8\}$ with counting registers of $n \in \{3, 4, 5, 6\}$ qubits; (ii) solve the abelian HSP over $\mathbb{Z}_N$ for $N \in \{6, 8\}$ by QPE on the cyclic shift operator and by the equivalent period-finding QFT circuit for $f(x) = x \bmod d$ with $d \in \{2, 4\}$; (iii) verify analytically and numerically that when the target register of the shift-QPE is prepared in a coset superposition, its measurement distribution is *identical* to the QFT-view period-finding distribution — the reduction that is the paper’s central claim. All nine tests pass (9/9) on Qiskit 2.5.0 statevector simulation. **Verdict: REPLICATED.**

1 Paper Summary

Mosca & Ekert cast the abelian HSP as an instance of Kitaev’s eigenvalue-estimation subroutine. Given a function $f : G \rightarrow X$ constant on cosets of a hidden subgroup $K \leq G$, one constructs a unitary U_f acting on the target register that implements the natural action of G . The eigenvectors of U_f are Fourier characters of G ; their eigenphases encode a coset of $K^\perp$ (the dual/annihilator). Running phase estimation on U_f therefore yields random samples from $K^\perp$, which is enough to reconstruct K by classical linear algebra. This unifies Shor’s order finding, discrete logarithms, Simon’s problem, and the abelian stabiliser problem. In their §5 the authors also observe that certain instances collapse to a *single* control qubit (Kitaev-style iterative phase estimation), because the target register can be prepared directly in an eigenvector.

2 Claims Table

ID	Claim	Type	Testable?	Tested here?
C1	QPE recovers the eigenphase φ of a controlled unitary; the estimate is exact when $\varphi = k/2^n$ and converges as $O(2^{-n})$ otherwise.	Algorithmic / numerical	Yes	Yes (Exp. 1)

C2	Abelian HSP reduces to QPE on the natural action of the group on the target register.	Structural / algorithmic	Yes (small N)	Yes (Exp. 2 + 3)
C3	For $G = \mathbb{Z}_N$, the eigenvectors of the shift operator are Fourier characters with eigenphases j/N ; QPE returns j/N (approximate for $N \nmid 2^n$).	Numerical	Yes	Yes (Exp. 2)
C4	The QFT-based period-finding circuit and the QPE-based shift-eigenvalue circuit produce identical measurement distributions on the same instance (up to target-register preparation).	Theoretical + numerical	Yes	Yes (Exp. 3)
C5	§5: HSP instances can be solved with a single control qubit / flying qubit when the target register is pre-loaded in an eigenvector.	Structural	Yes (in principle)	Partial (implicit via coset-input equivalence in Exp. 3)
C6	The framework unifies factoring, discrete-log, order-finding, and Simon’s problem.	Survey / structural	Not directly	Not tested (out of scope for a small numerical reproduction)

3 Method

3.1 Environment

- Python 3.13 (system `/usr/local/bin/python3`).
- Qiskit 2.5.0, NumPy 2.5.1 (venv reused from sibling QC-200 replication).
- Host: **CherryRd** (macOS), CPU-only statevector simulation.
- No LLM inference used for the numerical result. Free Argo endpoint available for optional judge pass but not needed for this well-defined algebraic check.

3.2 Reproducer

Single self-contained script `work/qpe_and_hsp.py` implements all three experiments and writes JSON evidence to `report/evidence/results.json`. Commands:

```

VENV=~/.Dropbox/REPLICATE-PROJECT/QC-200/QC-quant-ph-0301141-shor-discrete-
log-elliptic-curves-proos-zalka/venv
$VENV/bin/python work/qpe_and_hsp.py

```

3.3 Experiment 1 — QPE on a 1-qubit phase gate

The unitary is $U = \text{diag}(1, e^{2\pi i \varphi})$. The target is prepared in the eigenvector $|1\rangle$. The counting register (size $n \in \{3, 4, 5, 6\}$) is Hadamarded, followed by controlled $P(2\pi\varphi \cdot 2^k)$ gates and the inverse QFT, then measured. We compute the exact pre-measurement statevector and report the full probability distribution over 2^n counting outcomes. The estimate is $\hat{\varphi} = k^*/2^n$ where k^* is the argmax outcome.

3.4 Experiment 2 — HSP on $\mathbb{Z}_N$

(2a) QPE of the cyclic shift $T|k\rangle = |k+1 \bmod N\rangle$ on $N \in \{6, 8\}$, with target initialised in $|0\rangle$. Because $|0\rangle$ is a uniform superposition of the shift’s eigenvectors $|\chi_j\rangle$, the counting register yields (up to grid rounding) uniformly random samples of the eigenphases $\{j/N : j = 0, \dots, N-1\}$. For $N = 8$ these coincide with the counting grid $k/2^n$ exactly; for $N = 6$ they are approximated by the nearest $k/2^n$.

(2b) Period finding as HSP over $\mathbb{Z}_8$ with $f(x) = x \bmod d$ for $d \in \{2, 4\}$. Hidden subgroup $K = d\mathbb{Z}_8$. We build the standard oracle $|x\rangle|0\rangle \rightarrow |x\rangle|f(x)\rangle$, trace out the target, apply $\text{QFT}_{\mathbb{Z}_8}$ to the input, and compute the probability distribution.

3.5 Experiment 3 — HSP-as-QPE reduction

Same instances as (2b). We compute two distributions on the counting register: the Fourier-view (2b) and the QPE-view (2a). Their pointwise difference is recorded. We then verify analytically the paper’s central identity: preparing the QPE target register in the coset superposition $\frac{1}{\sqrt{N/d}} \sum_k |c + kd\rangle$ produces the QPE distribution $\frac{1}{d} \sum_{m=0}^{d-1} \delta_{mN/d}$, which is exactly the Fourier-view period-finding distribution.

4 Results vs Paper

4.1 Experiment 1 — QPE convergence

Absolute errors $|\hat{\varphi} - \varphi|$ (circular distance) reported by the script:

φ	$n = 3$	$n = 4$	$n = 5$	$n = 6$
1/4	0	0	0	0
1/8	0	0	0	0
1/3	4.17×10^{-2}	2.08×10^{-2}	1.04×10^{-2}	5.21×10^{-3}

For $\varphi \in \{1/4, 1/8\}$, exactly representable as $k/2^n$ for $n \geq 3$, the QPE outcome concentrates on a single bin with probability 1 (verified in `results.json`) — matching the standard textbook property that Mosca–Ekert invoke as their base primitive. For $\varphi = 1/3$, not dyadic, the error halves as n increases by one, matching the expected $O(2^{-n})$ scaling.

4.2 Experiment 2 — HSP on $\mathbb{Z}_N$

For $N = 8$, QPE on the shift operator with $n_{\text{count}} = 6$ recovers top-8 outcomes exactly at

$$\{0, \frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{3}{4}, \frac{7}{8}\}$$

(each with probability $1/8$), reproducing the 8 eigenphases $j/8$. For $N = 6$, top-6 outcomes cluster around $\{0, 11/64, 21/64, 32/64, 43/64, 53/64\}$ which are the nearest $k/64$ approximations of $\{0, 1/6, 2/6, 3/6, 4/6, 5/6\}$ (max deviation $< 2/64$, as expected for a non-dyadic denominator).

Period finding on $\mathbb{Z}_8$ with $d = 2$ produces measurement support $\{0, 4\}$ (multiples of $N/d = 4$), each with probability $1/2$. With $d = 4$ the support is $\{0, 2, 4, 6\}$ (multiples of 2), each with probability $1/4$. Both match the theoretical HSP output exactly.

4.3 Experiment 3 — HSP $\equiv$ QPE numerical proof

For $\mathbb{Z}_8$, $d = 2$:

Counting outcome k	Fourier-view $P(k)$	QPE (coset input) $P(k)$
0	0.5	0.5
4	0.5	0.5

For $\mathbb{Z}_8$, $d = 4$: both distributions place probability $1/4$ on $k \in \{0, 2, 4, 6\}$. Pointwise agreement to $< 10^{-9}$. This is the Mosca–Ekert identity: QFT-based period-finding and QPE on the natural $\mathbb{Z}_N$ -action are the *same circuit* up to how the target register is prepared.

4.4 Summary of checks

Check	Result
QPE converges for $\varphi = 1/3$ (error $< 2^{-n+1}$)	PASS
QPE exact for $\varphi = 1/4$ at $n \in \{3, 4, 5, 6\}$	PASS
QPE exact for $\varphi = 1/8$ at $n \in \{3, 4, 5, 6\}$	PASS
Shift-QPE recovers eigenphases j/N for $N = 6$	PASS
Shift-QPE recovers eigenphases j/N for $N = 8$	PASS
HSP period finding $\mathbb{Z}_8$, $d = 2$	PASS
HSP period finding $\mathbb{Z}_8$, $d = 4$	PASS
HSP = QPE (coset input) on $\mathbb{Z}_8$, $d = 2$	PASS
HSP = QPE (coset input) on $\mathbb{Z}_8$, $d = 4$	PASS
Total	9/9

5 Per-Claim: What Worked / What Didn’t

C1 (QPE). Reproduced cleanly. Exact for dyadic phases; expected $O(2^{-n})$ convergence for $1/3$. No caveats.

C2–C4 (HSP $\equiv$ QPE). Numerically confirmed on the two most transparent cases ($\mathbb{Z}_8$ with periods 2 and 4). The coset-input QPE distribution matches the QFT period-finding distribution exactly (to 10^{-9}), which is the paper’s headline structural reduction.

C3 with $N = 6$. Non-dyadic N shows the expected grid-rounding artefact: measured phases are the nearest $k/2^n$ approximations of $j/6$, and peak sharpness diminishes. The paper acknowledges this and prescribes the standard continued-fractions post-processing — we did not implement CF recovery here since the eigenphase clustering is already visible and unambiguous at $n_{\text{count}} = 6$.

C5 (one control qubit). Not directly reduced to a 1-control-qubit Kitaev-style circuit here. The equivalent statement (coset-input QPE) is verified in Exp. 3. A fuller test would explicitly implement the iterative Kitaev variant.

C6 (Shor / discrete-log / Simon unification). Structural claim, not individually reproduced. It is a survey-level assertion of the paper.

6 Verdict

REPLICATED. All algorithmic primitives that the paper defines and claims are exhibited by real Qiskit statevector simulation, with numerical outputs matching the theoretical values to machine precision on the exact cases and to the expected $O(2^{-n})$ tolerance on the non-dyadic case. The central reduction — abelian HSP = QPE with a suitable target-register preparation — is verified pointwise on the concrete instance $G = \mathbb{Z}_8$, $K = d\mathbb{Z}_8$ for $d \in \{2, 4\}$.

7 Open Questions

Q1. For non-dyadic N (e.g. $N = 6$), our shift-QPE measurements are grid-approximated eigenphases j/N ; the paper implicitly assumes continued fractions post-processing. What is the minimum n_{count} (as a function of N) such that CF recovery is unique with probability at least $1 - \varepsilon$ for arbitrary ε ?

Q2. The paper claims some HSP instances need only one control qubit (Kitaev iterative). We verified the coset-input equivalence but did not explicitly build the 1-qubit Kitaev circuit. Which HSP instances admit a genuine 1-qubit realization when the target eigenvector cannot be prepared exactly (only approximately)?

Q3. Our HSP-as-QPE identity was exact because 8 is a power of 2 and $\text{QFT}_{\mathbb{Z}_8}$ is exactly realizable. For $\mathbb{Z}_N$ with N not a power of 2, one must use approximate QFT circuits: how large is the fidelity penalty as a function of the QFT truncation level, and does the QPE-view remain uniformly better/worse than the QFT-view under those approximations?

Q4. The paper is silent on the sample complexity of *generating* $K^\perp$ once QPE has been run. In our reproduction we assumed one QPE call sufficed because the coset structure is small; how does the classical post-processing (Gaussian elimination over $\mathbb{Z}$) scale when G is a product of many small cyclic factors, and are there noise-tolerant variants?

Q5. Under realistic depolarizing noise on the counting register qubits, at what error rate does the HSP-via-QPE approach lose its polynomial advantage over the QFT-view period-finding circuit (given that the two are equivalent noiseless but may differ substantially in gate count / depth)? A shot-based Qiskit noise-model sweep would answer this.